CSE 107

# Object Oriented Programming

Term Final Question Bank


Jarin Subah (2405075)

Tahsin Ahmad (2405061)



Bangladesh University of Engineering and Technology

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Note:

1. This Question Bank (QB) contains all the questions from the past seven term final examinations. The questions have been systematically organized by topic for ease of study and practice.
2. If a question involves multiple topics, it has been categorized either under the topic that appears earlier in our topic list or under the topic that is most dominant in the question. A small number of questions could not be clearly assigned to any single topic; these have been placed in the “Miscellaneous” section.
3. Each question is labeled with one or more tags, allowing you to quickly locate and practice specific types of questions using your PDF reader’s search feature. There are three types of tags:
    - C++
    - Java
    - Theory

    The C++ and Java tags indicate that the question requires programming-based solutions, while Theory denotes descriptive or conceptual questions. Some questions may include a combination of these tags, depending on their nature.

4. This Question Bank has been carefully sorted, edited, and formatted to ensure clarity and usability. However, if you notice any errors or inconsistencies, please feel free to inform us. Your feedback is highly appreciated.

# OOP QB

## Philosophy of Object Oriented Programming

1. Explain the key features of object-oriented programming (OOP). How does a class differ from a structure in OOP? **[23-24 1a]** **[ Mark: 6+4 ]** **[ Tag: Theory ]**
2. Discuss the difference between a class and an object. Describe how C++ supports inheritance. **[21-22 1a]** **[Mark: 11]** **[ Tag: Theory ]**

## Classes & Objects

3. Write One C++ statement to implement each of the following actions.
    - Using the `new` operator, declare a pointer of the required type, allocate memory for a `char` variable, and assign it the value `E`.
    - Using the `new` operator, declare a pointer of the required type and allocate memory for an array of 5 `float` variables.
    - Using the `delete` operator, delete the entire integer array pointed to by `a`.
    - Using the `new` operator, declare a pointer of the required type and allocate memory for an object of class `A`, where no constructor is defined in class `A`. Assume that class `A` exists. **[22-23 3a]** **[ Mark: 12 ]** **[ Tag: C++ ]**
4. Identify all the problems in the following code segment.

```
static class StaticTest {
....static int x = 5, y;
....int z;
....{
........x = 5;
........z = 20;
....}
....static {
........y = x * 4;
........x = y;
....}
....int s1() {
........return x * y;
....}
....static void s2() {
........System.out.println("x = " + x);
........System.out.println("y = " + y);
........System.out.println("z = " + z);
....}
....public static void main(String[] args) {
........s1();
........System.out.println("x = " + x);
........System.out.println("z = " + z);
........s2();
....}
}
```

**[22-23 6a]** **[ Mark: 10 ]** **[ Tag: Java ]**

5. Using an example scenario, demonstrate the use of a static member variable and a static member function for a class. **[21-22 4b]** **[ Mark: 8 ]** **[ Tag: C++ ]**
6. Give an example scenario where you will use a static member variable for a class. **[Jan 21 8c]** **[ Mark: 10 ]** **[ Tag: C++, Java ]**

7. A `Student` class has a private variable to store ID, a constructor without parameter to initialize ID and a private static variable. Write the code segment for Class `Student` and to create an array of 120 objects of `Student` class in the main function using `new` operator. IDs of 120 `Student` objects are initialized in the constructor as 1, 2, ..., 120, respectively. **[Jan 20 1a]** **[ Mark: 10 ]** **[ Tag: C++ ]**
8. Describe a scenario when we cannot create an object of a class. **[Jan 20 3c]** **[ Mark: 7 ]** **[ Tag: Theory ]**
9. What are the possible problems if we use "call by value" for passing an object to a function? Explain with an example. How can we solve this problem? Explain in details with the example you provided. **[17-18 5a]** **[ Mark: 13 ]** **[ Tag: Theory ]**
10. Consider the following statement inside main function:
    `myClass ob[2] = {-1, -2};`
    Is this a valid statement? If yes, then what does it mean? If no, then what is the problem? **[17-18 5c]** **[ Mark: 5 ]** **[ Tag: C++]**
11. What is wrong with the following code segment?

    ```
    class myclass{
    ....int i,j;
    public:
    ....static void seti (int x) {this->i = x;}
    ....int geti(){return i;}
    };
    ```

    **[17-18 7c]** **[ Mark: 6 ]** **[ Tag: C++ ]**

## Inline Function

12. What is an inline function? How does it differ from an automatic inline function? Inline specifier is a request, not a command - explain. **[20-21 1b]** **[ Mark: 3+2+5=10 ]** **[ Tag: C++, Theory ]**
13. When and why will you make a function inline? When may a compiler ignore this specifier? When does automatic in-lining take place? **[17-18 6b]** **[ Mark: 7 ]** **[ Tag: C++, Theory ]**

## Friend Function

14. Consider the following class declaration-

    ```
    class Complex {
    ....int real;
    ....int imaginary;
    public:
    };
    ```

    - Write down an extractor method that generates the following form of input patterns:
      `Enter real part: 3`
      `Enter imaginary part: 8`
      Assume that both the real and the imaginary parts accept only positive and negative integers. The extractor method checks whether the data entered is valid or not. For invalid data, the method sets `failbit` to indicate improper input.
    - Write down an inserter method that produces output like 3+8i. **[23-24 3b]** **[ Mark: 7+3 ]** **[ Tag: C++ ]**
15. Design a program that defines a friend function which can access and modify the private variables of two different classes. **[22-23 2b]** **[ Mark: 15 ]** **[ Tag: C++ ]**
16. Write an example program to demonstrate the use of forward declaration of a class. **[21-22 3b]** **[ Mark: 11 ]** **[ Tag: C++ ]**
17. Evaluate the output of the following program:

    ```
    #include <iostream>
    using namespace std;
    ```

```
class circle{
....int r;
public:
....circle(int a = 0)
....{
........r = a;
....}
....void show() {
........cout << r << endl;
....}
....friend circle operator++(circle &ob);
};
circle operator++(circle &ob)
{
....ob.r++;
....return ob;
}
int main() {
....circle c1(10);
....++c1;
....c1.show();
....return 0;
}
```

[21-22 3d] [ Mark: 6 ] [ Tag: C++ ]

18. Consider the following class declaration-

```
class inventory {
....char item[40];
....int onhand;
....double cost;
public:
....inventory (char *i, int o, double c) {
........strcpy(item, i);
........onhand = o;
........cost = c;
....}
};
```

Write down the inserter and the extractor methods for the class and modify the class as necessary to work with the inserter and the extractor. [20-21 3c] [ Mark: 12 ] [ Tag: C++ ]

19. Consider the program shown below and answer the following:
    - Why an inserter function is implemented using a friend function?
    - What is the advantage of using "out" instead of "cout" inside the inserter function?

```
#include <iostream>
using namespace std;
class myclass {
....int x, y;
public:
....myclass (int x, int y) {this->x=x; this->y=y; }
....friend ostream &operator<<(ostream &out, myclass ob);
};
ostream &operator<<(ostream &out, myclass ob) {
....out << "x: " << ob.x << ", y: " << ob.y << endl;
....return out;
}
int main() {
```

```
....myclass ob (120, 130);
....cout << ob;
....return 0;
}
```

[Jan 21 7b] [ Mark: 15 ] [ Tag: C++ ]

20. "There are scenarios when a friend function is very useful for operator overloading"- support this statement for the following class `Rational` by writing a program that has to use friend function for operator overloading.

```
class Rational
{
....int n, d; // numerator and denominator
public:
....//Constructor(s) to initialize n and d
....Rational()
....{
........n=0;
........d=0;
....}
....Rational(int a, int b)
....{
........n=a;
........d=b;
....}
};
```

[Jan 20 2c] [ Mark: 15 ] [ Tag: C++ ]

21. "A friend function can be a global function not related to any particular class. A friend function can also be a member of another class." Explain this statement with appropriate examples. [17-18 5b] [ Mark: 12 ] [ Tag: C++, Theory ]

## Constructor & Destructor

22. Explain importance of using constructor and destructor in C++. Why and how does Java drop writing destructor in programs? [23-24 1b] [ Mark: 5+5 ] [ Tag: C++, Theory ]
23. Describe why constructor overloading is possible while destructor overloading is not allowed. [22-23 1a] [ Mark: 11 ] [ Tag: C++, Theory ]
24. Consider the program shown below and mention the line(s) where the copy constructor is called. What will be the output of this program?

```
#include <iostream>
using namespace std;
class MyClass {
public:
....int val;
....int multiplier;
....MyClass(int n, int m = 2) {
........val = n;
........multiplier = m;
....}
....MyClass(const MyClass& obj) {
........val = obj.val;
........multiplier = obj.multiplier;
....}
....void setValue(int v, int m = 5) {
........val = v;
........multiplier = m;
....}
....int calculateValue() {
```

```
........return val * multiplier;
....}
};
void functionC(MyClass* arr, int size) {
....for (int i = 0; i < size; ++i) {
........cout << arr[i].calculateValue() << endl;
....}
}
int main() {
....MyClass* arr = new MyClass[2]{MyClass(5, 2), MyClass(6)};
....arr[0] = arr[1];
....functionC(arr, 2);
....arr[0].setValue(15, 4);
....arr[1].setValue(30);
....cout << arr[0].calculateValue() << endl;
....cout << arr[1].calculateValue() << endl;
....delete[] arr;
....return 0;
}
```

[22-23 3c] [ Mark: 13 ] [ Tag: C++ ]

25. Differentiate between `new` operator and `malloc()` function? Design "Class A" so that the following code inside the main function does not produce any error:

```
int main()
{
....A obj(10);
....A *array = new A[10];
....return 0;
}
```

[21-22 1c] [ Mark: 13 ] [ Tag: C++ ]

26. Consider the following class:

```
class strtype {
....char *p;
....int len;
public:
....strtype(char *s) {
........len = strlen(s) + 1;
........p = new char[len];
........strcpy(p, s);
....}
....~strtype() {
........delete [] p;
....}
};
```

Identify the problem that may arise if we want to execute the following instructions from the main function:

```
strtype s1 ("LEVEL"), s2("TERM");
s1 = s2;
```

Without changing the constructor and the destructor function, write the code to solve the problem. [21-22 2c] [ Mark: 15 ] [ Tag: C++ ]

27. What does a default constructor (generated by the compiler) perform? What are the scenarios when a copy constructor is called? [Jan 21 5c] [ Mark: 10 ] [ Tag: C++, Theory ]

28. Explain what a default copy constructor generated by the compiler do and briefly describe a circumstance where this will cause a problem? [Jan 20 1b] [ Mark: 10 ] [ Tag: C++, Theory ]

29. What is the output of the following program:

```
#include<iostream>
using namespace std;
class Point{
private:
....int x;
....int y;
public:
....Point(int i=0, int j=0); //Normal Constructor
....Point (const Point &t); // Copy Constructor
};
Point::Point(int i, int j) {
....x=i; y=j;
....cout << "Normal Constructor called\\n";
}
Point::Point(const Point &t) {
....x=t.x; y=t.y;
....cout << "Copy Constructor called\\n";
}
int main()
{
....Point *t1, t2;
....t1 = new Point(10, 15);
....t2 = new Point(*t1);
....Point t3 = *t1;
....Point t4;
....t4 = t3;
....return 0;
}
```

[17-18 7b] [ Mark: 9 ] [ Tag: C++ ]

## Operator Overloading

30. If you overload the assignment operator, should you return an object of the same type as the class it is overloaded for? Justify your answer. [22-23 2a] [ Mark: 10 ] [ Tag: C++, Theory ]
31. Consider the program shown below. You are not allowed to modify the given code, but you can add additional code to the `Coordinator` class to produce the following output (in bold):
**x coordinate: 7**
**y coordinate: 10**
You do not need to rewrite the full program. Simply write the additional code you want to add to the `Coordinator` class.

```
#include <iostream>
using namespace std;
class Coordinator {
....int x, y;
public:
....Coordinator(int x_val = 0, int y_val = 0) {
........x = x_val;
........y = y_val;
....}
....int getX() { return x; }
....int getY(){ return y; }
};
int main() {
....Coordinator cObj(5, 8);
```

```
....cObj += 2;
....cout << "x coordinate: " << cObj.getX() << endl;
....cout << "y coordinate: " << cObj.getY() << endl;
....return 0;
}
```

[22-23 2c] [ Mark: 10 ] [ Tag: C++ ]

32. Design your custom inserter in the program below to generate the following output:
`x: 120, y: 130`

```
#include <iostream>
using namespace std;
class myclass {
....int x, y;
public:
....myclass(int x, int y) {
........this->x = x;
........this->y = y;
....}
};
int main(){
....myclass ob(120, 130);
....cout << ob;
....return 0;
}
```

[21-22 2b] [ Mark: 10 ] [ Tag: C++ ]

33. Consider the program shown below. Though you are not allowed to modify the given code, you can add additional code. Write necessary code that you want to add in `Photo` and/or `Frame` class to produce the following output (in bold):
**Frame Width: 7**
**Frame Length: 10**
You do not need to write the full program. Only write the code that you want to add and mention where you want to add it.

```
#include <iostream>
using namespace std;
class Photo{
....int width, length;
public:
....Photo () {width=0; length=0;};
....Photo (int a, int b) {width=a; length=b;};
....int getwidth(){return width;}
....int getlength(){return length;}
};
class Frame {
....int width, length;
public:
....Frame(){width=0; length=0;};
....Frame (int a, int b) {width=a; length=b;};
....int getwidth(){return width;}
....int getlength(){return length;}
};
int main()
{
....Photo P(5, 8);
....Frame F;
....F = P + 2;
```

```
....cout << "Frame Width: " << F.getwidth() << endl;
....cout << "Frame Length: " << F.getlength() << endl;
....return 0;
}
```

[Jan 21 6a] [ Mark: 20 ] [ Tag: C++ ]

34. Consider the program shown below. Overload the prefix operator for `Fraction` class using a friend function to produce the following output (in bold):

**x/y: 4/8**

You do not need to write the full program. Only write the overloaded function.

```
#include <iostream>
using namespace std;
class Fraction {
....int x, y;
public:
....Fraction (int a, int b) {x=a; y=b;};
....int getx(){return x;}
....int gety(){return y;}
};
int main()
{
....Fraction n(3, 7);
....++n;
....cout << "x/y: " << n.getx() << "/" << n.gety() << endl;
....return 0;
}
```

[Jan 21 8a] [ Mark: 13 ] [ Tag: C++ ]

35. If you overload `operator+`, you must return an object of the same type as your parameters. Do you agree? Justify your answer. [Jan 20 2b] [ Mark: 7 ] [ Tag: C++, Theory ]

36. Given the following class declaration `Date`, overload the binary minus operator so that an expression like `ob1-ob2` will give the difference between the two `Date` objects in number of days, where `ob1` and `ob2` are objects of `Date` class. For simplicity of calculation assume that all months have 30 days. Also overload left shift operator so that an expression like `cout<<ob` will print day-month-year (e.g., 17-1-2021), where `ob` is an object of `Date` class. Include a `main()` function where these operators will be used.

```
class Date {
....int day, month, year;
public:
....Date(int m, int d, int y) {
........day=d;
........month=m;
........year=y;
....}
};
```

[Jan 20 4c] [ Mark: 17 ] [ Tag: C++ ]

37. Consider the following class:

```
class Complex
{
public:
....int real, imag;
....Complex (int r=0, int i=0);
};
```

Consider three objects of `Complex` class as `comp1`, `comp2` and `comp3`. Now write necessary functions so that the following statements can be executed in the main function.

- `comp3 = comp1 + comp2;`
  The "real" value of `comp3` should be the added value of `comp1` and `comp2`'s "real" value. The "imag" value of `comp3` should be the added value of `comp1` and `comp2`'s "imag" value. `comp1` and `comp2` should not change.
- `comp3 = comp1 + 10;`
  The "real" value of `comp3` should be the "real" value of `comp1` increased by 10. The "imag" value of `comp3` should be the "imag" value of `comp1`. `comp1` and `comp2` should not change.
- `comp3 = 10 + comp1;`
  The "imag" value of `comp3` should be the "imag" value of `comp1` increased by 10. The "real" value of `comp3` should be the "real" value of `comp1`. `comp1` and `comp2` should not change.
- `comp1++;`
  Both the "real" and "imag" value of `comp1` should be the increased by 1. This statement should behave like the usual postfix increment. That means the increment must occur after the operand is evaluated.
- `++comp1;`
  Both the "real" and "imag" value of `comp1` should be the increased by 1. This statement should behave like the usual prefix increment. That means the increment must occur before the operand is evaluated.
- `cout << comp1;`
  This should be print the complex number in an "a + bi" manner. For example, if the "real" and "imag" value of `comp1` is 2 and 3 respectively, then this statement should print "2 + 3i". If the "real" and "imag" value of `comp1` is 4 and -5 respectively, then this statement should print "4 - 5i".
- `comp3 = comp1 (5);`
  The "real" value of `comp3` should be the added value of `comp1`'s "real" value and 5. The "imag" value of `comp3` should be the added value of `comp1`'s "imag" value and 5. `comp1` should not change. **[17-18 6a] [ Mark: 25 ] [ Tag: C++ ]**

38. Which of the operators can have default arguments when they are overloaded? **[17-18 7e] [ Mark: 3 ] [ Tag: C++, Theory ]**
39. Why the overloaded assignment operator (=) function should return the `this` pointer? **[17-18 8d] [ Mark: 5 ] [ Tag: C++, Theory ]**

## Type Conversion

40. Differentiate between early binding and late binding with examples. Demonstrate `const_cast` in case of `const` method with example. **[23-24 3c] [ Mark: 4+4 ] [ Tag: C++ ]**
41. Write a program to demonstrate a user-defined conversion function that converts a class object to a built-in data type. **[22-23 4a] [ Mark: 9 ] [ Tag: C++ ]**

## Inheritance

42. What is erasure in terms of generic class? Consider the following main function of C++ and write down associated generic classes where the class `Gen` inherits the class `NonGen` and the class `Gen2D` inherits the class `Gen`.

```
int main() {
....NonGen ob(10);
....Gen<char> ob1(20, 'A');
....Gen2D<double, char *> ob2(40, 60.5, "Gen2Object");
....ob.show();
....ob1.show();
....ob2.show();
....cout << ob1.getob() << endl;
....cout << ob2.getob() << endl;
```

```
....return 0;
}
```

Where the output of the program is as follows:

```
NonGen: 10
Gen: A
Gen2D: Gen2Object
A
Gen2 Object
```

[23-24 4a] [ Mark: 15 ] [ Tag: C++ ]

43. Demonstrate the access of private, public, and protected members of a base class in a derived class and in the main function, when the derived class inherits the base class using private inheritance. Provide an example program. [22-23 1b] [ Mark: 12 ] [ Tag: C++, Theory ]
44. Demonstrate the order of execution of constructor functions for multilevel inheritance with an example program. [21-22 1b] [ Mark: 11 ] [ Tag: C++ ]
45. Is it possible that a derived class inherits more than one copy of the base class? If yes, show the scenario; how can you avoid it? [Jan 21 6b] [ Mark: 15 ] [ Tag: C++ ]
46. Write an example program to explain the order of execution of constructor functions for multilevel inheritance and multiple inheritance. [Jan 20 3a] [ Mark: 15 ] [ Tag: C++ ]

## Function Overloading

47. Describe the differences between function overloading and function overriding. [21-22 2a] [ Mark: 10 ] [ Tag: Theory ]
48. Consider the following overloaded functions and develop the equivalent function using default arguments:

    ```
    void f2() { int a = 0, b = 0; ... }
    void f2(int a) { int b = 0; ... }
    void f2(int a, int b) { ... }
    ```

    [21-22 3c] [ Mark: 6 ] [ Tag: C++ ]
49. "Generic functions are similar to overloaded functions except that they are more restrictive"- Do you agree? Justify your answer. [21-22 4a] [ Mark: 5 ] [ Tag: C++, Theory ]
50. Two overloaded functions are declared as follows-

    ```
    void space(char ch) {
    }
    void space(char ch, int count) {
    }
    ```

    Write down the `main()` function that declares two variables for storing the addresses of the above two functions and make function calls using those variables. What problem is observed if the second function is declared as follows-

    ```
    void space(char ch, int count = 0)
    }
    ```

    [20-21 2a] [ Mark: 7 ] [ Tag: C++ ]
51. Write a short note on "function overloading". [Jan 21 5a] [ Mark: 10 ] [ Tag: Theory ]
52. Differentiate between function overloading and function overriding. [Jan 20 2a] [ Mark: 8 ] [ Tag: Theory ]

## Virtual Function

53. What is the diamond problem? State with example how the diamond problem can be solved using virtual inheritance. Differentiate between virtual function and virtual inheritance. [23-24 3a] [ Mark: 4+3+3 ] [ Tag: C++, Theory ]

54. Design a program to show the use of virtual functions in achieving runtime polymorphism. [22-23 1c] [ Mark: 12 ] [ Tag: C++, Theory ]
55. Describe the use of a virtual base class. Explain why it is not possible to create an object of an abstract class? [21-22 3a] [ Mark: 12 ] [ Tag: C++, Theory ]
56. Differentiate between early binding and late binding with appropriate examples. [20-21 3a] [ Mark: 8 ] [ Tag: C++, Theory ]
57. Explain runtime polymorphism with an example program. [Jan 21 7a] [ Mark: 20 ] [ Tag: Theory ]
58. Differentiate between compile time polymorphism and run time polymorphism. [Jan 20 3b] [ Mark: 8 ] [ Tag: Theory ]
59. Can constructor or destructor functions be declared as virtual? Explain briefly. [17-18 5d] [ Mark: 5 ] [ Tag: C++, Theory ]
60. What is early binding and late binding in C++? Explain with examples. [17-18 7a] [ Mark: 12 ] [ Tag: C++, Theory ]
61. Why virtual base classes are used? Give an example of a class hierarchy where we must use virtual base class to avoid ambiguity. You do not need to write code. [17-18 8b] [ Mark: 9 ] [ Tag: C++, Theory ]

## Function Template

62. Design a template function `myfunc` which takes two numbers (both of them are integers or both of them are real numbers) and an operator as a character ( `+`, `-`, `*` or any other value). The function prints the summation, subtraction and product of the two numbers for `+`, `-` and `*`, respectively. For any other value in operator, the function prints Invalid Input. A sample main function and its corresponding output is shown below:

```
#include <iostream>
using namespace std;
int main()
{
....myfunc(10.5, 4.3, '+');
....myfunc(4, 8, '*');
....myfunc(10.5, 4.5, '-');
....myfunc(3, 6, 'a');
....return 0;
}
```

Output:

```
14.8
32
6
Invalid Input
```

[21-22 4c] [ Mark: 10 ] [ Tag: C++ ]

## Class Template

63. What is template class? Write down hierarchy of template classes and associated 8-bit character-based classes used in C++. [23-24 4d] [ Mark: 5 ] [ Tag: C++, Theory ]
64. Consider the code snippet of program shown below. The expected output of the program is:
`1.1 2.2 3.3`
`20 25`
Now write the additional code that you want to add such that the program produces the expected output.

```
#include <iostream>
using namespace std;
int main(){
....myclass<double> d1(1.1);
....myclass<double,2> d2(1.1);
```

```
....myclass<double,3> d3(1.1);
....cout << d1.getx() << " " << d2.getx() << " " << d3.getx() << "\n";
....myclass<int,4> i1(5);
....myclass<int> i2(5);
....cout << i1.getx() << " " << i2.getx();
....return 0;
}
```

[22-23 4b] [ Mark: 10 ] [ Tag: C++ ]

65. Write a class `myCalculator` using class templates to show the result of addition, subtraction, multiplication and division between two numbers. The numbers can be integer, float or double. The main function and a sample output are given below.

```
int main()
{
....myCalculator<int> intCalc (7, 4);
....myCalculator<float> floatCalc (3.4, 6.2);
....cout << "Int results:" << endl;
....intCalc.displayResult();
....cout << endl << "Float results:" << endl;
....floatCalc.displayResult();
....return 0;
}
```

```
Output:
Int results:
Numbers are: 7 and 4.
Addition is: 11
Subtraction is: 3
Product is: 28
Division is: 1

Float results:
Numbers are: 3.4 and 6.2.
Addition is: 9.6
Subtraction is: -2.8
Product is: 21.08
Division is: 0.548387
```

[17-18 8a] [ Mark: 15 ] [ Tag: C++ ]

## STL

66. What is STL? Explain different fundamental items of STL. [23-24 4b] [ Mark: 5 ] [ Tag: C++, Theory ]
67. Consider that a list container contains the following data:
`20 16 12 8 4 3 6 9 12 15`
Using STL algorithms, write down a C++ program that will perform the following operations on the list and show the results after each operation-
    - Count how many times 12 appears in the list
    - Count how many even numbers are there in the list
    - Remove all appearance of 12 from the list and store the new list into a separate container without changing the contents of existing list
    - Reverse the original list

- Replace all numbers of the original list by its squares using `transform` and `iterator`
  The program shall include necessary declarations and function definitions. [23-24 4c] [ Mark: 10 ] [ Tag: C++ ]

68. Write a program that demonstrates the use of the class `map` of C++ standard template library (STL) and fulfills the following requirements:
    - Create a class `st_record` that stores then name and cgpa of a student as types `string` and `double`, respectively. Add functions to the class as necessary to complete your code.
    - In the `main` function, create an object of STL class `map` named `result` to store the records of students against their student numbers (as `int`), i.e., the types of `<key,value>` pair for the map object will be `<int, st_record>`.
    - Inset some dummy data (records of at least four students) in the map object `result`.
    - Traverse the map object `result`, find the student with the highest cgpa and print his information- student number, name and cgpa. Assume there will only be one student with the highest cgpa. [22-23 4c] [ Mark: 16 ] [ Tag: C++ ]
69. Discuss different types of containers that are available in C++ Standard Template Library (STL). Describe their differences with an example class for each type of container. [21-22 4d] [ Mark: 12 ] [ Tag: C++, Theory ]
70. Consider that the `map <stdId, student>` is used in an object-oriented programming to store the records of the students, where `stdID` (i.e., student identification number) is unique for each student. A object "student" contains student ID, student name, address, mobile number and CGPA. The class "student" contains three member functions - a constructor, a getter for student ID and a `display()` function showing all of the above information about a student. Write down the class "student". Also write down a C++ program that
    - takes information about a student from the keyboard and inserts a pair into the map
    - scan `stdId` from keyboard buffer and print all information abut that student from its associated object using map. Make necessary assumptions. [20-21 4a] [ Mark: 20 ] [ Tag: C++ ]

## Java

71. Consider the following code segment:

```
public class Main {
....public static void main (String[] args) {
........int a = minmax ("min", 2, 1, 6, 4, 5); // a = 1
........int b = minmax ("min", 3, 0, 6); // b = 0
........int c = minmax ("max", 1, 2, 6, 5); // c = 6
........int d = minmax ("max", 1, 3, 7); // d = 7
....}
}
```

    Write the above `minmax` function in Java. You are allowed to write only one `minmax` function. [22-23 6b, Jan 20 5c] [ Mark: 10 ] [ Tag: Java ]
72. Write 2 (two) different ways of creating the following array in Java. [20-21 5b] [ Mark: 10 ] [ Tag: Java ]

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 6 | 7 | 8 | 9 | |
| 10 | 11 | 12 | | |
| 13 | 14 | | | |
| 15 | | | | |

73.
    13. What is the difference between the following two declarations in Java?
    14. `int c[][],x`
    15. `int [][]c,x`
    Write 2 (two) different ways of creating the following array in Java.

| | | | |
|---|---|---|---|
| 0 | | | |
| 1 | 2 | | |
| 3 | 4 | 5 | |
| 6 | 7 | 8 | 9 |

**[Jan 20 5a] [ Mark: 10 ] [ Tag: Java ]**

74. What are the problems with the following Java code?

```
public class TestStatic {
....static int a = 3, b;
....int c;
....{
........c = 10;
....}
....static {
........b = a * 4;
........c = b;
....}
....int f2() {
........return a * b;
....}
....static void f1(int x) {
........System.out.println("x = " + x);
........System.out.println("a = " + a);
........System.out.println("b = " + b);
........System.out.println("c = " + c);
....}
....public static void main (String[] args) {
........f1(42);
........System.out.println("b = " + b);
........System.out.println("Area = " + f2());
....}
}
```

**[Jan 20 5b] [ Mark: 10 ] [ Tag: Java ]**

## Nested & Inner Class

75. List all the problems in the following Java code. Write Java code to fix only the problems in the main method of the `Test` class.

```
class X {
....int a = 1;
....void display() {
........Y obY = new Y();
........obY.display();
........Z obZ = new Z();
........obZ.display();
....}
....void showNested() {
........System.out.println(b);
........System.out.println(c);
....}
....class Y {
........int b = 2;
........void display() {
............System.out.println(a);
```

```
............System.out.println(b);
........}
....}
....static class Z {
........int c = 3;
........void display() {
............System.out.println(a);
............System.out.println(c);
........}
....}
}
class Test {
....public static void main(String[] args) {
........X obX = new X();
........obX.display();
........obX.showNested();
........Y obY = new Y();
........obY.display();
........Z obZ = new Z();
........obZ.display();
....}
}
```

[23-24 6a] [ Mark: 10 ] [ Tag: Java ]

76. Write Java codes to fix the problems (if any) in the main method of the following code segment.

```
class P {
....int x = 1;
....class Q {
........int y = 2;
....}
....static class R {
........int z = 3;
....}
}
public class NestedClassTest {
....public static void main(String[] args) {
........P p = new P();
........p.print();
........p.show();
........Q q = new Q();
........q.print();
........R r = new R();
........r.print();
....}
}
```

[20-21 6d] [ Mark: 5 ] [ Tag: Java ]

77. What are two types of nested classes? What are the differences between them? List all the problems in the following Java code. Write Java codes to fix only the problems in the main method of `Test` class.

```
class X {
....int a = 1;
....void display () {
........Y obY = new Y ();
........obY.display();
........Z obZ = new Z ();
........obZ.display();
....}
```

```
....void showNested () {
........System.out.println(b);
........System.out.println(c);
....}
....class Y {
........int b = 2;
........void display () {
............System.out.println(a);
............System.out.println(b);
........}
....}
....static class Z {
........int c = 3;
........void display () {
............System.out.println(a);
............System.out.println(c);
........}
....}
}
public class Test {
....public static void main (String[] args) {
........X obX = new X ();
........obX.display();
........obX.showNested();
........Y obY = new Y();
........obY.display();
........Z obZ = new Z ();
........obZ.display();
....}
}
```

**[Jan 21 4b] [ Mark: 20 ] [ Tag: Java ]**

78. What is an inner class? Point out the problem in the following code segment (if any).

```
class Outer {
....int outer_x=5;
....void test() {
........Inner inner = new Inner();
........inner.display();
....}
....class Inner {
.......int z=60;
........void display() {
............System.out.println(outer_x);
........}
....}
....void showz() {
........System.out.println(z);
....}
}
```

**[17-18 1c] [ Mark: 9 ] [ Tag: Java ]**

## String, String Buffer & String Builder

79. What will be the output of the following Java code:

```
class Q1 {
....public static void main(String[] args) {
........String s1 = "Hello";
........String s2 = new String("Hello");
........String s3 = "Hello";
........s2.intern();
........System.out.println(s1 == s2);
........System.out.println(s1 == s3);
........System.out.println(s2 == s3);
........final String s0 = "Hello";
........final String t0 = "World";
........String t1 = s0 + t0;
........String t2 = new String("HelloWorld");
........String t3 = "HelloWorld";
........t3.intern();
........System.out.println(t1 == t2);
........System.out.println(t1 == t3);
........System.out.println(t2 == t3);
....}
}
```

What is the problem with the following code segment with respect to efficiency? How would you rewrite this for efficiency, considering only one thread is used?

```
String result = "";
for (int i = 0; i < 10000; i++) {
....result += i;
}
```

**[23-24 6c] [ Mark: 15 ] [ Tag: Java ]**

80. Describe 3 (three) different ways to convert an `int` value to a `String`. Explain the major differences between an `ArrayList` and a `Vector`. **[21-22 7a] [ Mark: 10 ] [ Tag: Java, C++, Theory ]**
81. Consider your own `StringTokenizer` class - `MyStringTokenizer` with the following methods:
    - `MyStringTokenizer (String str, String delimiter)` - the only constructor that takes the main String and the delimiter for tokenize
    - `int countTokens ()` - returns total number of tokens
    - `boolean hasMoreTokens ()` - returns false if no more token left, true otherwise
    - `String nextToken ( )` - returns the next token
    Write complete Java code for your `MyStringTokenizer` class. You can add necessary instance variables. You are not allowed to use the original `StringTokenizer` class, but you can use methods of the `String` class. **[20-21 6a] [ Mark: 10 ] [ Tag: Java ]**
82. Write three ways to declare an object of string class. **[Jan 20 4b] [ Mark: 8 ] [ Tag: Java ]**
83. Write a Java program that takes a `String` as input and prints the number of tokens (the delimiter is space) in that String as well as the tokens.

```
Sample Input: This is a test
Output:
4
This
is
a
test
```

## Interface

84. What is the problem with the code below? Write two different ways to fix the problem. **[Jan 21 1c] [ Mark: 5 ] [ Tag: Java ]**

```
interface A {
....default void f () {
........System.out.println("A's f");
....}
}
interface B {
....default void f () {
........System.out.println("B's f");
....}
}
public class C implements A, B {
}
```

85. Consider the following code segment. Now complete `MyClass` with minimum code for successful compilation. **[17-18 2b] [ Mark: 9 ] [ Tag: Java ]**

```
interface Int1 (
....void f1();
....void f2();
}
interface Int2 extends Int1{
....void f3();
}
class MyClass implements Int2{
....// Complete this class
}
public class Test{
....public static void main (String[] args) {
........MyClass my = new MyClass();
........my.f1();
........my.f2();
........my.f3();
....}
}
```

## Abstract Class

86. Consider the following code segment.

```
interface X1 {
....void m1();
....static void m2();
....default void m3();
}
interface X2 {
....private void m4();
....void m5();
}
abstract class A1 implements X1 {
....final void m6();
....abstract void m7();
```

```
}
class A2 extends A1 implements X2 {
....// your code
}
```

- Outline the errors in the provided code (if any) and fix them without removing any access modifiers.
- After the fix, demonstrate the minimum code in writing for class `A2` so that it compiles successfully. You can't define `A2` as abstract. [23-24 5a] [ Mark: 10 ] [ Tag: Java ]

87. What will be the output of the following code:

```
abstract class X {
....X() {
........System.out.println("X Constructor");
........draw();
....}
....abstract void draw();
....abstract void process(int x);
....void process(Object o) {
........System.out.println("X process Object");
....}
}
class Y extends X {
....int y = 5;
....Y() {
........System.out.println("Y Constructor");
....}
....void draw() {
........System.out.println("Y with y = " + y);
....}
....void process(int x) {
........System.out.println("Y process int");
....}
....void process(String s) {
........System.out.println("Y process String");
....}
}
class Test {
....public static void main(String[] args) {
........X obj = new Y();
........obj.process(10);
........obj.process("hello");
........obj.process((Object)"hello");
....}
}
```

[23-24 6b] [ Mark: 10 ] [ Tag: Java ]

88. Consider the following code segment.

```
interface i1 {
....default void f1();
....void f2();
....static void f3();
}
interface i2 {
....void f4();
....private void f5();
}
abstract class c1 implements i1 {
```

```
....abstract void f6();
....final void f7() {
....}
}
class c2 extends c1 implements i2 {
....// your code
}
```

- Outline the errors in the provided code (if any) and fix them without removing any access modifiers.
- After the fix, demonstrate the minimum code in writing for class `c2` so that it compiles successfully. You can't define `c2` as abstract. **[22-23 5a] [ Mark: 10 ] [ Tag: Java ]**

89. Suppose there are seven methods defined as follows: `void f1(), void f2(), void f3(), void f4(), void f5(), void f6() and void f7()`. There are also three interfaces named `i1, i2, and i3` and one abstract class named `A`. There is a class named `MyClass` that needs to be forced to implement all the above seven methods, where you must maintain the following constraints:
    - Each interface can define at most two methods.
    - The abstract class `A` can only implement one interface and can define at most one abstract method.
    - The class `MyClass` can't implement any interface.
      Determine and then write the complete Java code to achieve the above scenario. **[22-23 5b, 20-21 7a] [ Mark: 10 ] [ Tag: Java ]**
90. Explain three uses of the `final` keyword with short examples. Describe the fundamental differences between an interface and an abstract class. **[21-22 5a] [ Mark: 10 ] [ Tag: Java, Theory ]**
91. Consider the following code segment:

```
interface x1 {
....void f1();
....static void f2() {}
....default void f3() {}
}
interface x2 {
....void f4();
....private void f5() {}
}
abstract class A implements x1 {
....final void f6() {}
....abstract void f7();
}
public class B extends A implements x2 {
....// your code
}
```

Demonstrate successful compilation of class B by writing the minimum code. You can't define class B as abstract. **[21-22 5b] [ Mark: 10 ] [ Tag: Java ]**

92. Explain abstraction with an appropriate example. Write down the properties of an abstract class and an abstract method. **[20-21 1a] [ Mark: 5+3+2=10 ] [ Tag: Java, Theory ]**
93. You are given the following Java code with the main method.

```
class Animal {
....private String name;
....private int age;
....public Animal(String name, int age) {
........this.name = name;
........this.age = age;
....}
}
class TestAnimal {
```

```
....public static void main(String[] args) {
........Bird alex = new Albatross("Alex", 39);
........Mammal spot = new Dog("Spot", 7);
........Mammal fred = new FruitBat("Fred", 10);
........Reptile steph = new EasternBrownSnake("Steph", 12);
........Fish bruce = new GreatWhiteShark("Bruce", 40);
........Arachnid charlotte = new RedBackSpider("Charlotte", 1);
........Mammal perry = new Platypus ("Perry", 5);
........Mammal bob = new Human ("Bob", 20);
........Animal[] animals = {alex, spot, fred, steph, bruce, charlotte, perry, bob};
....}
}
```

Write Java code for the necessary classes to support the main method where you can't create any object of the `Animal` class. You can add/modify code to the `Animal` class if required, but you are not allowed to change the main method. **[20-21 6b]** **[ Mark: 10 ]** **[ Tag: Java ]**

94. Consider the following code segment. Write the minimum code in class `B` for successful compilation. You can't define `B` as abstract.

```
interface x1 {
....void f1();
....static void f2() {}
....default void f3() {}
}
interface x2 {
....void f4();
....private void f5() {}
}
abstract class A implements x1 {
....final void f6() {}
....abstract void f7();
}
public class B extends A implements x2 {
....// your code
}
```

**[Jan 21 1a]** **[ Mark: 10 ]** **[ Tag: Java ]**

95. Consider the following code segment:

```
interface i1 {
....default void f1() {
....}
....static void f2() {
....}
....void f3();
}
interface i2 {
....void f4();
....void f5();
}
abstract class c1 implements i1 {
....abstract void f6();
....final void f7() {
....}
}
class c2 extends c1 implements i2 {
....// your code
}
```

Write minimum code in class `c2` for successful compilation. You can't define `c2` as abstract. [Jan 20 6a] [ Mark: 10 ] [ Tag: Java ]

96. Suppose there are 8 methods defined as follows:
`void m1()`, `void m2()`, `void m3()`, `void m4()`, `void m5()`, `void m6()`, `void m7()`, and `void m8()`.
There are also 4 interfaces named as: `interface x1`, `x2`, `x3` and `x4`.
There is also a class named `MyClass` that needs to be forced to implement all the above 8 methods, where you have to maintain the following constraints:
    1. Each interface can define at most 2 methods.
    2. The class `MyClass` can only implement 1 interface.
    Write Java code for `MyClass` to achieve the above scenario. [Jan 20 6b] [ Mark: 10 ] [ Tag: Java ]
97. What is an abstract class? Write the restrictions imposed on an abstract class. Point out and fix the problems in the following code snippet.

```
final class X{
....final void show() {
........System.out.println("Printed from a method");
....}
}
class Y extends X{
....void show() {
........System.out.println("In subclass method");
....}
}
```

[17-18 2a] [ Mark: 11 ] [ Tag: Java ]

## Object Class

98. Consider the following code segment, where the expected output is given as comments.

```
class Book {
....private String title;
....private String author;
....private int year;
....private int edition;
....public Book(String title, String author, int year, int edition) {
........this.title = title;
........this.author = author;
........this.year = year;
........this.edition = edition;
....}
....public static void main(String[] args) {
........Book b1 = new Book("1984", "George Orwell", 1949, 1);
........Book b2 = new Book("1984", "George Orwell", 1949, 2);
........System.out.println(b1);
........// Book (1984, George Orwell, 1949, 1)
........System.out.println(b1 == b2);
........// false
........System.out.println(b1.equals(b2));
........// true
........HashSet<Book> set = new HashSet<>();
........set.add(b1);
........set.add(b2);
........System.out.println(set.size());
........// 1
```

```
....}
}
```

Design the complete Java code for the `Book` class to achieve the expected output. Please note that `HashSet` is a Java Collections class that stores unique elements. **[23-24 5c] [ Mark: 15 ] [ Tag: Java ]**

99. Consider the following code segment:

```
class Point {
....private int x, y;
....Point(int x, int y) {
........this.x = x;
........this.y = y;
....}
}
class Circle {
....Point center;
....int radius;
....Circle(int x, int y, int r) {
........center = new Point(x, y);
........this.radius = r;
....}
}
public class Test {
....public static void main(String[] args) {
........Circle c1 = new Circle(10, 20, 5);
........Circle c2 = new Circle(10, 20, 5);
........System.out.println(c1);
........System.out.println(c2);
........System.out.println(c1.equals(c2));
........HashMap<Circle, String> m = new HashMap<>();
........m.put(c1, "I am a Circle");
........System.out.println(m.get(c2));
....}
}
```

The expected output of the main method:

```
Circle: center (10, 20), radius: 5
Circle: center (10, 20), radius: 5
true
I am a Circle
```

Construct the `Point` and `Circle` class to achieve the expected output. Please note that the `Point` class's getter methods for `x` and `y` are private. **[21-22 5c] [ Mark: 15 ] [ Tag: Java ]**

100. Consider the following code segment. The expected output of the above code segment is given as comments. Complete the `Movie` class to achieve the expected output. **[20-21 6c] [ Mark: 10 ] [ Tag: Java ]**

```
public class Movie {
....private String title;
....private int releaseYear;
....private String productionCompany;
....private int runningTime;
....public Movie(String title, int releaseYear, String productionCompany, int runningTime) {
........this.title = title;
........this.releaseYear = releaseYear;
```

```
........this.productionCompany = productionCompany;
........this.runningTime = runningTime;
....}
....public static void main(String[] args) {
........Movie m1 = new Movie("The Lord of the Rings", 2001, "New Line Cinema", 178);
........Movie m2 = new Movie("The Lord of the Rings", 2001, "WingNut Films", 178);
........System.out.println(m1 == m2); // false
........System.out.println(m1.equals(m2)); // true
........HashMap map = new HashMap();
........map.put(m1, 93);
........System.out.println(map.get(m2)); // 93
....}
}
```

101. Consider the following code segment:

```
public class Institute {
....private int eiin;
....private int shift;
....private int version;
....private int group;
....public Institute (int eiin, int shift, int version, int group) {
........this.eiin = eiin;
........this.shift = shift;
........this.version = version;
........this.group = group;
....}
....public static void main (String [] args) {
........Institute i1 = new Institute (135790, 1, 1, 0);
........Institute i2 = new Institute (135790, 1, 1, 0);
........System.out.println(i1.equals(i2));
........HashMap map = new HashMap();
........map.put (i1, 100);
........System.out.println(map.get(i2));
....}
}
```

The expected output of the above code:

```
true
100
```

Complete the `Institute` class to achieve the expected output. **[Jan 20 6c]** **[ Mark: 10 ]** **[ Tag: Java ]**

## Exception Handling

102. What is the problem in the following C++ program? Rewrite the program using exception handling to overcome the problem? **[23-24 3d]** **[ Mark: 2+5 ]** **[ Tag: C++ ]**

```
#include <iostream>
using namespace std;
int divide() {
....int d = 0;
....int a = 10 / d;
....return a;
```

```
}
int main() {
....cout << divide() << endl;
....return 0;
}
```

103. Consider a Java program that copies a text file named "src.txt" to another text file named "dst.txt" line by line. Implement this Java program using try-with-resources to close the files automatically. Please note that you are not allowed to throw `Exception` anywhere in your program. **[23-24 8b] [ Mark: 10 ] [ Tag: Java ]**
104. Write the name of the five keywords in Java dedicated to exception handling. Consider a class named `BankAccount` with the following information:
`double balance` - the balance of the account
`void init (double amount)` - sets the amount as the initial balance
`void debit (double amount)` - subtracts the amount from the balance
`void credit (double amount)` - adds the amount directly to the balance The amount provided in the above methods can't be negative. The balance of the account can't also be negative.
Write Java code for the following:

- A custom exception ( `InvalidAmountException` ) that triggers when the given amount is negative.
- A custom exception ( `InsufficientBalanceException` ) that triggers when the balance goes negative.
- The complete `Account` class to trigger these exceptions when needed.
You don't need to write any main method. **[22-23 6c, Jan 21 1b] [ Mark: 15, 20] [ Tag: Java ]**

105. Consider a Java program that copies a binary file named "src.mkv" to another binary file named "dst.mkv" using chunks of 8 KB (8092 bytes). Implement this Java program using try with resources to close the files automatically. Please note that you are not allowed to throw `Exception` anywhere in your program. **[21-22 6b] [ Mark: 10 ] [ Tag: Java ]**
106. A "queue" is a data-structure in which a newly entered element is placed in the back of the queue ; however, elements are served or going out of the queue from the front. Write down a generic class named "queue" where all types of data such as integer, character, string, double and object of any kind can be stored and queued. The queue will support following member functions: `push()` , `pop()` , `size()` .
Write down the codes for the necessary exception handling during performing different operation on the queue. **[20-21 4b] [ Mark: 15 ] [ Tag: Java ]**
107. Consider a class named `Inventory` with the following information:

- `double stock` - the stock of the inventory
- `double minStock` - the minimum stock of the inventory
- `void setInitialStock (double amount)` - this method sets the amount as the initial stock
- `void addToInventory (double amount)` - this method adds the amount to the stock
- `void removeFromInventory (double amount)` - this method subtracts the amount from the stock
The amount provided in the above four methods can't be negative. The stock of the inventory can't also be below the minimum stock.
Write Java code for the following:
- A custom exception named `InvalidStockAmountException` with appropriate parameters that triggers when the given amount is negative and shows the message 'The given stock amount can't be negative'.
- A custom exception named `InvalidInventoryStockException` with appropriate parameters that triggers when the stock goes below the minimum stock and shows the message 'The stock can't be less than the minimum stock'.
- The complete `Inventory` class to trigger these exceptions when needed. You don't need to write any main method.
**[20-21 7b] [ Mark: 10 ] [ Tag: Java ]**
.

## Multithreading

108. Consider the following code:

```
class TestClass {
....public void f0() {
........synchronized (this) {
............for (int i = 0; i < 5; i++)
................System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
....public void f1() {
........synchronized (TestClass.class) {
............for (int i = 0; i < 5; i++) {
................System.out.println(Thread.currentThread().getName() + " " + i);
............}
........}
....}
....public synchronized static void f2() {
........for (int i = 0; i < 5; i++) {
............System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
....public synchronized static void f3() {
........for (int i = 0; i < 5; i++) {
............System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
}
class SynchronizedTest {
....public static void main(String[] args) {
........new Thread(new TestClass()::f1, "T1").start();
........new Thread(TestClass::f2, "T2").start();
........new Thread(TestClass::f3, "T3").start();
....}
}
```

- Determine and then write down all the possible outputs of the above code segment, assuming that thread T1 will start executing first.
- If we create two different threads executing methods `f0` and `f1`, will they run in parallel? Why or why not? **[23-24 5b] [ Mark: 10 ] [ Tag: Java ]**

109. What is the problem with the following code segment? Write 2 (two) different ways to fix it. **[23-24 7a] [ Mark: 10 ] [ Tag: Java ]**

```
class MyThread implements Runnable {
....List<String> list;
....Thread t;
....public MyThread(List<String> list) {
........this.list = list;
........t = new Thread(this);
........t.start();
....}
....public void run() {
........for (int i = 1; i <= 10000; i++) {
............list.add(String.valueOf(i));
```

```
........}
....}
}
public class TestSyncArrayList {
....public static void main(String[] args) throws Exception {
........List<String> list = new ArrayList<>();
........MyThread myThread1 = new MyThread(list);
........MyThread myThread2 = new MyThread(list);
........myThread1.t.join();
........myThread2.t.join();
........System.out.println(list.size());
....}
}
```

110. Complete the following Java code that finds the summation of a 10000-element integer array using an array of 10 threads in parallel. You cannot write any more classes, but you can add variables, constructors, and other methods. **[23-24 7c] [ Mark: 15 ] [ Tag: Java ]**

```
class ParallelSum implements Runnable {
....// your code here
}
public class Main {
....public static void main(String[] args) {
........java.util.Random random = new java.util.Random();
........int[] numbers = new int[10000];
........for (int i = 0; i < numbers.length; i++) {
............numbers[i] = random.nextInt(10000);
........}
........ParallelSum[] parallelSum = new ParallelSum[10];
........// your code here
....}
}
```

111. You are given the following class:

```
class ThreadTest {
....public void f0() {
........for (int i = 1; i < 500; i++) {
............System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
....public synchronized void f1() {
........for (int i = 1; i < 1000; i++) {
............System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
....public synchronized void f2() {
........for (int i = 1; i < 1500; i++) {
............System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
....public synchronized static void fs1() {
........for (int i = 1; i < 2000; i++) {
............System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
```

```
....public synchronized static void fs2() {
........for (int i = 1; i < 2500; i++) {
............System.out.println(Thread.currentThread().getName() + " " + i);
........}
....}
}
```

You have the following restrictions:
R1: You are not allowed to change `ThreadTest` class
R2: You can create only one object of `ThreadTest` class
Now, you have to create the following threads from the main method:
T1: This thread will execute the `f1` method
T2: This thread will execute the execute `f2` method
T3: This thread will execute the execute `f0` method
T4: This thread will execute the execute `fs1` method
T5: This thread will execute the execute `fs2` method
Design the Java code for the main method. Now, compose the answer to the following questions:

- Which of the threads will run in parallel and why?
- Which of the threads will not run in parallel and why? Can you make them parallel with the other threads by relaxing restriction R2? If yes, what changes do you have to make in your code? **[22-23 5c] [ Mark: 15 ] [ Tag: Java ]**

112. You are given a task to find the total count of prime numbers in a randomly generated array of 10000 integers with values between 1 and 5000. You are asked to divide the work equally among five different threads. Write complete Java code to compute the total count of prime numbers by dividing the work equally among five threads. The main thread will wait for the five threads to finish and only print the final count. You can use the following code segment to generate a random number between 1 and 5000. **[22-23 7c] [ Mark: 15 ] [ Tag: Java ]**

```
java.util.Random random = new Random();
int x = random.nextInt(5000)+1;
```

113. Describe two different ways of creating threads in Java with short examples. Which one is better and why? Explain the difference between a synchronized method and a synchronized statement. **[21-22 6a] [ Mark: 10 ] [ Tag: Java, Theory ]**

114. The following Java code (incomplete) is required to find the minimum of a 1000 integers array using an array of 5 threads in parallel.

```
class ParallelMin implements Runnable {
....// your code
}
class Main {
....public static void main(String[] args) {
........Random random = new Random();
........int[] numbers = new int[1000];
........for (int i = 0; i < numbers.length; i++) {
............numbers[i] = random.nextInt();
........}
........ParallelMin [] parallelMin = new ParallelMin[5];
........// your code
....}
}
```

You need to construct the complete Java code. You cannot write any more classes, but you can add variables, constructors, and other methods. **[21-22 6c]** **[ Mark: 15 ]** **[ Tag: Java ]**

115. With Java threads, it is very easy to parallelize computations. Suppose you are in a job interview with Oracle and the interviewer asks you to write Java code to find out summation of 1 to 10000. You can't use any mathematical equation; you can only use loops. But you are asked to divide the work equally among 5 different threads. Write complete Java code to compute the summation of 1 to 10000 by dividing the work equally among 5 different threads. The main thread will wait for the 5 threads to finish and will only print the final summation. **[20-21 7c]** **[ Mark: 10 ]** **[ Tag: Java]**

116. Consider the following `SharedCounter` class:

```
class SharedCounter {
....private int counter;
....SharedCounter () {
........this.counter = 0;
....}
....private void increment () {
........this.counter++;
....}
....public int get() {
........return this.counter;
....}
....public void count () {
........for (int i = 1; i <= 1000; i++) {
............increment();
........}
....}
}
```

Explain with code how you can fix the `SharedCounter` class so that multiple threads can correctly operate on it. You don't need to write the whole class. **[20-21 7d]** **[ Mark: 5 ]** **[ Tag: Java ]**

117. You are given the following class:

```
class TestClass {
....public void f1() {
........for (int i = 0; i < 5; i++) {
............System.out.println(i);
............try {
................Thread.sleep(1000);
............} catch (InterruptedException e) {
................System.out.println(e);
............}
........}
....}
....public static void f2() {
........for (int i = 0; i < 5; i++) {
............System.out.println(i);
............try {
................Thread.sleep(2000);
............} catch (InterruptedException e) {
................System.out.println(e);
............}
........}
....}
}
```

You have to create two threads from the main method. The first thread will execute the `f1` method, and the second thread will execute the `f2` method. Write Java code for the main method. You are not allowed to change the `TestClass`; that means `TestClass` can't extend/implement any class/interface. **[Jan 21 3a] [ Mark: 10 ] [ Tag: Java ]**

118. Mr. P and Mr. C worked in the same warehouse. Their work is to put and get products in a shared inventory of size 4. Mr. P puts a single product in the inventory, and Mr. C gets a single product from the inventory at a time. Mr. P cannot put a product if the inventory is full with 4 products. Mr. C cannot get a product if the inventory contains no product. Mr. P can put a product in the inventory if any one of the inventory places is empty. Mr. C can get a product from the inventory if any one of the inventory places is full with a product. Write complete Java code to solve the problem mentioned above using the concepts of inter-thread communication. You have to use `wait` and `notify` / `notifyAll`, and you are not allowed to use `BlockingQueue`. **[Jan 21 3b] [ Mark: 20 ] [ Tag: Java ]**

119. What are two common ways to achieve synchronization between Threads? Which one gives you more control over the synchronization? **[Jan 21 3c] [ Mark: 5 ] [ Tag: Java, Theory ]**

120. With Java threads, it is very easy to parallelize computations. Suppose you are in a job interview and the interviewer asks you to write Java code to find out summation of 1 to 10000000. You can't use any simple equation; you can only use loops. But you are asked to divide the work equally among 10 different threads. Write complete Java code to compute the summation of 1 to 10000000 by dividing the work equally among 10 different threads. The main thread will wait for the 10 threads to finish and will only print the final summation. **[Jan 20 7a] [ Mark: 10 ] [ Tag: Java ]**

121. Consider two entities, the Writer and the Reader, who share a common buffer that is a single character. The Writer's job is to generate English letters circularly from A to Z. The Reader's job is to read and display them in the output. The Writer can't write new letter if the Reader does not read the already written letter. The Reader can't read if the Writer does not write any new letter. Write a java code to solve the above-mentioned problem using the concept of inter thread communication. You can use `wait` / `notifyAll` or `ArrayBlockingQueue`. **[Jan 20 7b] [ Mark: 10 ] [ Tag: Java ]**

122. Write three different ways to create Threads in Java with short code examples. What is the difference between synchronized method and synchronized statement? **[Jan 20 7c] [ Mark: 10 ] [ Tag: Java, Theory ]**

123. Briefly describe the methods that Java uses for interthread communication. **[17-18 2c] [ Mark: 9 ] [ Tag: Java, Theory ]**

124. What are the advantages of multi-threading over process based multitasking? **[17-18 2d] [ Mark: 6 ] [ Tag: Java, Theory ]**

## Generics & Collections

125. Consider the following code segment and its output:

```
public class MyMaxTest {
....public static void main(String[] args) {
........Long[] n1 = {10L, 20L, 30L, 40L, 50L};
........MyMax<Long> mx1 = new MyMax<>(n1);
........System.out.println(mx1.max());
........Float[] n2 = {50.0f, 20.0f, 40.0f, 10.0f, 30.0f};
........MyMax<Float> mx2 = new MyMax<>(n2);
........System.out.println(mx2.max());
........Double[] n3 = {50.0, 20.0, 40.0, 10.0, 30.0};
........MyMax<Double> mx3 = new MyMax<>(n3);
........System.out.println(mx3.max());
........System.out.println(mx1.sameMax1(mx2));
........System.out.println(mx2.sameMax2(mx3));
....}
}
```

Output:

```
50.0
50.0
```

```
50.0
true
true
```

Implement the `MyMax` class to achieve the provided output for the `main` method with the following restrictions:

- `MyMax` class can only be created for Java number classes.
- The implementation of methods `sameMax1` and `sameMax2` will be different. **[23-24 8a] [ Mark: 10 ] [ Tag: Java ]**

126. Consider the following class where getter methods are available for all attributes:

```
public class Employee {
....private int id;
....private String name;
....private int age;
....Employee(int id, String name, int age) {
........this.id = id; this.name = name;
........this.age = age;
....}
}
```

Write Java code for the following:

- Declare an `ArrayList` named `list` that can store a list of `Employees`, and add the following employees to the list:

| Id | Name | Age |
|---|---|---|
| 1 | Rodgers | 10 |
| 2 | Bradley | 25 |
| 3 | Chambers | 15 |
| 4 | Calvin | 10 |
| 5 | Quinn | 30 |
| 6 | Mccoy | 15 |

- Rewrite the following two code segments using Java Stream:

```
// Segment 1
List<String> names = new ArrayList<>();
for (int i = 0; i < list.size(); i++) {
....Employee e = list.get(i);
....names.add(e.getName());
}
```

```
// Segment 2
List<Employee> employees = new ArrayList<>();
for (int i = 0; i < list.size(); i++) {
....Employee e = list.get(i);
....if (e.getAge() <= 20) employees.add(e);
}
```

- Sort the list based on the ascending order of the age. If the age is the same, then sort based on the ascending order of the name. You can't use your own sorting technique. However, you can change the `Employee` class if necessary.

[23-24 8c] [ Mark: 15 ] [ Tag Java: ]

127. Write two differences between `ArrayList` and `Vector`.
Consider the following class where getter methods are available for all the attributes:

```
public class Employee {
....private int id;
....private String name;
....private int age;
}
```

Suppose there is an `ArrayList` of `Employee` named `listEmployees`. Write necessary Java code so that when `Collections.sort(listEmployees)` is called, the employees are sorted by ascending order of `id`, and if the `id` matches, then by lexicographic order of `name`, and if both match, then by ascending order of `age`. [22-23 7a] [ Mark: 10 ] [ Tag: Java ]
128. Write three differences between `Hashtable` and `HashMap`. Consider the following information on different IPL player's weekly salaries:

| Name | Salary |
|---|---|
| Pat Cummins | 17500000 |
| Kane Williamson | 16000000 |
| MS Dhoni | null |

Write Java code for the following:

- Create an appropriate Hash-based collection to store the information and then store the above three information of players.
- Iterate through all the keys of the above Hash-based collection and print their values without any hardcoding.
- Increase Pat Cummin's salary by 500,000. [22-23 7b] [ Mark: 10 ] [ Tag: Java ]

129. Write a generic interface named `iQueue` with methods `enqueue`, `dequeue`, `size`, and `isEmpty`. Then, write a generic class `Queue` that implements the `iQueue` interface. Note that the `iQueue` interface only supports numeric types. [22-23 8a, Jan 20 8b] [ Mark: 10 ] [ Tag: Java ]
130. Consider the following code segment and its output:

```
public class MyStatsTest {
....public static void main(String[] args) {
........Integer[] n1 = {1, 2, 3, 4, 5};
........List<Integer> list1 = Arrays.asList(n1);
........MyStats<Integer> s1 = new MyStats<>(list1);
........System.out.println(s1.average());
........Long[] n2 = {5l, 2l, 4l, 1l, 3l};
........List<Long> list2 = Arrays.asList(n2);
........MyStats<Long> s2 = new MyStats<>(list2);
........System.out.println(s2.average());
........Double[] n3 = {5.0, 4.0, 3.0, 2.0, 1.0};
........List<Double> list3 = Arrays.asList(n3);
........MyStats<Double> s3 = new MyStats<>(list3);
........System.out.println(s3.average());
........System.out.println(s1.sameAvg1(s2));
........System.out.println(s2.sameAvg2(s3));
```

```
....}
}
```

Output: `3.0`, `3.0`, `3.0`, `true`, `true`.

Implement the `MyStats` class to achieve the provided output for the `main` method with the following restrictions:

- `MyStats` class can only be created for Java number classes.
- The implementation of methods `sameAvg1` and `sameAvg2` will be different. **[22-23 8b]** **[ Mark: 10 ]** **[ Tag: Java ]**

131. Consider the following code segment and its output.

```
public class Q {
....public static void main (String [] args) {
........Integer [] n1 = {10, 20, 30, 40, 50};
........MyAvg<Integer> s1 = new MyAvg<>(n1);
........System.out.println(s1.average());
........Integer [] n2 = {50, 20, 40, 10, 30};
........MyAvg<Integer> s2 = new MyAvg<>(n2);
........System.out.println(s2.average());
........System.out.println(s1.sameAvg(s2));
........Double [] n3 = {50.0, 40.0, 30.0, 20.0, 10.0};
........MyAvg<Double> s3 = new MyAvg<>(n3);
........System.out.println(s3.average());
........System.out.println(s2.sameAvgAny(s3));
....}
}
```

The output of the main method is as follows:

```
30.0
30.0
true
30.0
true
```

Implement the `MyAvg` class to achieve the provided output for the main method. Please note that the `MyAvg` class can only be created for Java number classes. **[21-22 7b]** **[ Mark: 10 ]** **[ Tag: Java ]**

132. Consider the following class where getter methods are available for all attributes:

```
public class Employee {
....private int id;
....private String name;
....private int age;
....Employee (int id, String name, int age) {
........this.id = id;
........this.name = name;
........this.age = age;
....}
}
```

Develop Java code for the following:

133. Consider the following class where getter methods are available for all attributes:

```
public class Employee {
....private int id;
....private String name;
....private int age;
....Employee (int id, String name, int age) {
........this.id = id;
........this.name = name;
........this.age = age;
}
}
```

Develop Java code for the following:
- Declare an `ArrayList` named `employeeList` that can store a list of employees and add the following employees to the list.

| Id | Name | Age |
|---|---|---|
| 1 | Eric | 10 |
| 2 | Brown | 25 |
| 3 | Ben | 15 |
| 4 | Peter | 10 |
| 5 | Jack | 30 |
| 6 | Stewart | 15 |

- Store only the name of the employee whose age is greater than 15 in an `ArrayList` named `nameList`.
- Sort the list based on the ascending order of the age. If the age is the same, then sort based on the ascending order of the name. You can't use your own sorting technique. However, you can change the `Employee` class if necessary. **[21-22 7c] [ Mark: 15 ] [ Tag: Java ]**

134. What are the differences between `ArrayList` and `Vector`? Consider the following class:

```
public class Fruit {
....private String name;
....private int quantity;
}
```

Suppose there is an `ArrayList` of `Fruit` named `listFruits`. Write necessary Java code so that when `Collections.sort(listFruits)` is called, the products are sorted by descending order of quantity and if quantity matches then by lexicographic order of their name. **[20-21 8a] [ Mark: 10 ] [ Tag: Java ]**

135. Write a generic interface named `iStack` with methods `push`, `pop` and `isEmpty`. Then write a generic class `Stack` that implements the `iStack` interface. Please note that `iStack` interface only supports numeric types. **[20-21 8b] [ Mark: 10 ] [ Tag: Java ]**

136. Consider the following code segment and its output.

```
public class Q{
....public static void main (String[] args) {
........Integer []n1={10,20,30,40,50};
........MyStats<Integer> s1 = new MyStats<>(n1);
........System.out.println(s1.average());
........Integer [] n2={50,20,40,10,30};
........MyStats <Integer>s2=new MyStats<>(n2);
```

```
........System.out.println(s2.average());
........System.out.println(s1.sameAvg(s2));
........Double [] n3 = { 50.0, 40.0, 30.0, 20.0, 10.0 };
........MyStats<Double> s3 = new MyStats<>(n3);
........System.out.println(s3.average());
........System.out.println(s2.sameAvgAny(s3));
....}
}
```

The output of the main method is as follows:

```
30.0
30.0
true
30.0
true
```

Write complete Java code for the `MyStats` class. Please note that `MyStats` class can only be created for Java number classes. **[Jan 21 2a]** **[ Mark: 10 ]** **[ Tag: Java ]**

137. Consider the following class.

```
public class Person {
....private int id;
....private String name;
....private int age;
....Person (int id, String name, int age) {
........this.id = id;
........this.name = name;
........this.age = age;
....}
....public int getId () {
........return id;
....}
....public String getName () {
........return name;
....}
....public int getAge () {
........return age;
....}
}
```

Write Java code for the following:

- Declare an `ArrayList` named `list` that can store a list of `Persons` and add the following persons to the list:

| Id | Name | Age |
|---|---|---|
| 1 | Rodgers | 10 |
| 2 | Bradley | 25 |
| 3 | Chambers | 15 |
| 4 | Calvin | 10 |
| 5 | Quinn | 30 |
| 6 | Mccoy | 15 |

- Rewrite the following two code segments using Java Stream:

```
// Segment 1
List<String> names = new ArrayList<>();
for (int i = 0; i < list.size(); i++) {
....Person mc = list.get(i);
....names.add(mc.getName());
}

// Segment 2
List<Person> persons = new ArrayList<>();
for (int i = 0; i < list.size(); i++) {
....Person mc = list.get(i);
....if (mc.getAge() <= 20) {
........persons.add(mc);
....}
}
```

- Sort the list based on the ascending order of the age. If the age is the same, then sort based on ascending order of the name. You can't use your own sorting technique. However, you can change the `Person` class if necessary. **[Jan 21 2b] [ Mark: 20 ] [ Tag: Java ]**

138. What are the differences between `ArrayList` and `Vector`? Which collection class is good for manipulating data? **[Jan 21 2c] [ Mark: 5 ] [ Tag: Theory ]**
139. Consider the following class:

```
class Product {
....private String name;
....private double price;
....Product (String name, double price) {
........this.name = name;
........this.price = price;
....}
....public String getName () {
........return this.name;
....}
....public double getPrice () {
........return this.price;
....}
}
```

Write Java code for the following:

- Define an ArrayList named `myProducts` that can store a list of `Product`.
- Generate 4 random `Product` with names 'A' to 'D' and random price and add them to `myProducts`.
- Sort `myProducts` based on Product's name in ascending order. You can't use your own sorting techniques. You can change the `Product` class if necessary. **[Jan 20 8c] [ Mark: 10 ] [ Tag: Java ]**

140. Write the general syntax for (i) declaring a generic class (ii) declaring a reference to a generic class and instance creation. **[17-18 1b] [ Mark: 10 ] [ Tag: Java ]**
141. Write three differences between `Hashmap` and `Hashtable`. Consider the following monthly salary of different cricketers:

| Name | Salary (Tk.) |
|---|---|
| Sakib Al Hasan | 3,00,000 |
| Mashrafe Bin Mortaza | null |
| Mushfiqur Rahim | 2,50,000 |

Now write Java code for the following:

- Create an appropriate Hash-based collection to store the above information and then store the above information.
- Increase Sakib Al Hasan's salary by 50,000 Tk. **[17-18 4b]** **[ Mark: 16 ]** **[ Tag: Java ]**

## Server Socket

142. A Fibonacci server can accept multiple Fibonacci clients. The client program will send two integers to the server. The server will calculate the number of Fibonacci numbers between the two integers and send it to the client. The client will show the number as output. The server and client will only communicate through objects. Write a complete Java client-server program to handle the above scenario. **[22-23 8c]** **[ Mark: 15 ]** **[ Tag: Java ]**
143. A server program can accept multiple clients. The client program will send two integers to the server. The server will calculate the number of prime numbers between the two integers and send it to the client. The client will show the number as output. Develop a complete Java server-client program to handle the above scenario. **[21-22 8c]** **[ Mark: 15 ]** **[ Tag: Java ]**
144. Suppose a client wants to connect to a server on IP address 192.168.1.1 and port 44444. Just draw the scenario of operations needed for the successful connection establishment and data transfer between the server and client. You do not need to write any code. What are the names of the classes generally used to read and write objects in networking? What restriction do they impose on the class? **[20-21 8c]** **[ Mark: 10 ]** **[ Tag: Java, Theory ]**
145. Briefly describe the types of TCP Sockets. Write a Java Program that opens a connection to a "whois" port (port 43) on the InterNIC server `whois.internic.net`, sends the command-line argument (a website address) over the socket and then prints the data that is returned. **[17-18 3a]** **[ Mark: 19 ]** **[ Tag: Java ]**

## Type Wrapper

146. Describe boxing and unboxing in Java. Give examples with primitive `int` and `Integer`. Explain the difference between boxing/unboxing and autoboxing/unboxing with code. When you shouldn't use autoboxing/unboxing? **[21-22 8a]** **[ Mark: 10 ]** **[ Tag: Java, Theory ]**
147. Consider the following code segment:

```
public class AutoBoxingUnboxingDemo {
....static int f (Integer v) {
........return v;
....}
....public static void main (String args[]) {
........Integer a = f(100);
........Integer iOb, iOb2;
........int i;
........iOb = 100;
........++iOb;
........iOb2 = iOb + (iOb / 3);
........i = iOb + (iOb / 3);
........Integer intOb = 100;
........Double doubleOb = 98.6;
........doubleOb = doubleOb + intOb;
```

```
....}
}
```

List with explanation all the auto-boxing and auto-unboxing performed in the main method. **[Jan 21 4a]** **[ Mark: 10 ]** **[ Tag: Java ]**
148. What do you mean by auto-boxing and auto-unboxing? Explain with code examples. When you shouldn't use them? Write three differences between `Hashtable` and `HashMap`? **[Jan 20 8a]** **[ Mark: 10 ]** **[ Tag: Java, Theory ]**
149. What is a type wrapper? Name any three type wrappers. What are type wrappers needed? **[17-18 3b]** **[ Mark: 10 ]** **[ Tag: Java, Theory ]**
150. What are boxing and unboxing? Give an example for each of them. **[17-18 4a]** **[ Mark: 10 ]** **[ Tag: Java, Theory ]**

## Miscellaneous

151. Why is it important to write custom manipulator in C++ programming? Write down a C++ program that generates following output:

```
Enter a hexadecimal number: c2d
You entered: *****0XC2D
```

The program must contain two custom manipulators named `hex_input` and `setup`. The manipulator `hex_input` takes a number in hexadecimal form and the manipulator `setup` displays the number on screen in uppercase taking 10 characters, showing base and displaying `*` as shown instead of blanks. **[23-24 1c]** **[ Mark: 5+10 ]** **[ Tag: C++ ]**
152. Consider the following class declaration:

```
class Coord {
....int x, y;
public:
};
```

Create a `Rectangle` class that uses the `Coord` class as follows:

```
class Rectangle {
....Coord p1, p2, p3, p4;
....char *name;
public:
};
```

Answer the following questions and write down methods as instructed below:

- Write down a constructor of `Coord` class accepts only positive values of `x` and `y` with no single `x` or `y` coordinate larger than 200.
- Write down a constructor of `Rectangle` class that accepts four points and checks whether they form a rectangle or not. A rectangle has right angles at each of its corner (and checking of any three corners are sufficient). Write a static function `isRightAngle()` that accepts three points and checks whether a right angle is formed at the middle point or not?
- Write down four member functions of `Rectangle` class for computing the length, the width, the perimeter and the area of the rectangle. The length is the larger of the two dimensions. Take care to reuse methods where possible.
- Include a predicate function `square` that determines whether the rectangle is a square or not.
- Write down a clone constructor that makes a clone of rectangle.
- Write down necessary conversion methods for executing the following program codes:

```
int value = rect; //returns the area of rect
string name = rect;
```

Where `rect` is a rectangle type object.

- Briefly state the problem while executing the following code- `r1 = r2;` Where both `r1` and `r2` are rectangle type object. Overwrite the assignment operator to overcome the problem. **[23-24 2]** **[ Mark: 35 ]** **[ Tag: C++ ]**

153. Create a Java enum named `HttpStatus` to represent the following HTTP status codes:
 `OK (200, "OK")`, `NOT_FOUND (404, "Not Found")`, `INTERNAL_ERROR (500, "Internal Server Error")`
 Each enum constant should store two fields: an `int` status code and a `String` message. Your enum should also implement the following methods:

- `isError()`: Returns `false` for `OK`, and `true` for both `NOT_FOUND` and `INTERNAL_ERROR`
- `toString()`: Returns a string in the format `"200 OK"`, `"404 Not Found"`, `"500 Internal Server Error"`
- `fromCode(int code)` (static method): Takes an `int` code and returns the corresponding `HttpStatus` constant. If the code doesn't match any defined status, return `null`. **[23-24 7b]** **[ Mark: 10 ]** **[ Tag: Java ]**

154. "If a function returns a reference of any type, then it can be used both as an l-value and r-value"- show with an example program. **[22-23 3b]** **[ Mark: 10 ]** **[ Tag: Theory ]**
155. Consider the following countries, their number of FIFA World Cup titles, and the year of the last title in the FIFA World Cup:

| Country | Number of Titles | Year of the Last Title |
|---|---|---|
| Argentina | 3 | 2022 |
| France | 2 | 2018 |
| Uruguay | 2 | 1950 |
| England | 1 | 1966 |
| Spain | 1 | 2010 |

Implement the following using Java code for the above scenario:

- Declare a single Enumeration named `FWCWinners` for the above countries.
- Write Java code to print all the countries with their number of titles and the year of the last title.
- Write Java code to print the most recent winner (you can't simply write Argentina). **[21-22 8b]** **[ Mark: 10 ]** **[ Tag: Java ]**

156. Imagine a tollbooth at a bridge. Cars passing by the booth are expected to pay a 50-cent toll. Mostly they do, but sometimes a car goes by without paying. The tollbooth keeps track of the number of cars that have gone by, and of the total amount of money collected.
 Model this tollbooth with a class called `tollbooth`. The three data items are a list of string to hold the registration numbers of the cars passing through the bridge, a type `unsigned int` to hold the total number of cars, and a type `double` to hold the total amount of money collected. A constructor initializes both of these to 0. A member function called `payingCar()` adds the car registration number in the list, increments the car total and adds 0.50 to the cash total. Another function, called `nopayCar()`, adds the car registration number, increments the car total but adds nothing to the cash total. Finally, a member function called `display()` displays the list of cars' registration numbers and the two totals. Make appropriate member functions `const`. Use proper naming convention and setter-getter functions. **[20-21 1c]** **[ Mark: 15 ]** **[ Tag: C++, Java ]**

157. Let be a vector where and are real numbers as well as and represent unique vectors along x-axis, y-axis and z-axis. There is a description for each point of the vector and hence each object has a pointer to point the description. Declare a class named `vector` that has members and/or friends to fulfill the following requirements. Don't use friend function if it can be implemented using member function. **[20-21 2b] [ Mark: 28 ] [ Tag: C++ ]**

| Requirement | Sample Code | Explanation |
|---|---|---|
| Creating instance | `Vector ob(12.0, 5.0, 2.0), ob1(3.0, 2.0, 5.0, "The peak point"), ob2;` | Use default arguments to set the vector point at the origin with null description when no parameter is given. Write appropriate constructor and copy constructor. |
| Addition of two complex number objects | `ob1 = ob2 + ob3;` | The values in each dimension will be added separately, i.e., the x values of `ob2` and `ob3` will be added together to give x value of `ob1` and so on. Descriptions of `ob2` and `ob3` will be concatenated by adding a space between them to give the description of `ob1`. |
| Incrementing a complex number | `ob = ob1 += ob2;` | Overwrite `=` and `+=` operators. |
| Adding a real number to an object | `ob2 = ob1 + 100.0;` `ob2 = 100.0 + ob1;` | The real number will be added to each value of the object, i.e., each value of x, y and z will be incremented by the amount of the real number. |

158. Write a C++ program that emulates the DOS "filesize" command and returns the size in bytes of a program entered on the command line. Invoke the program with one command-line argument - the file name - like this: `filesize program.ext`. In the program, check that the user has typed the correct number of command-line arguments, and that the file specified can be opened. **[20-21 3b] [ Mark: 7 ] [ Tag: C++ ]**

159. Write down a manipulator named "setup" that will generate the following output against the statement of `cout<<setup<<123.45678;`
123.456%%%
**[20-21 3d] [ Mark: 8 ] [ Tag: C++ ]**

160. Identify the problems in the following code segment and write the complete code after fixing them. **[20-21 5a] [ Mark: 10 ] [ Tag: Java ]**

```
class A {
....var x;
....void getX() {
........return x;
....}
....void setX(int x) {
........x = x;
....}
....public static void main(String[] args) {
........A [] a = new A[10];
........for (int i = 0; i < a.length(); i++) {
............a[i].setX(i);
........}
........for (int i = 0; i < a.length(); i++) {
............System.out.println(a[i].getX());
........}
....}
}
```

161. A singleton class is a class that can have only one object at a time. After the first time, if we try to instantiate a singleton class, the new variable also points to the first object created. Write a complete Java code to create a

singleton class named `MySingleton`, where the single instance can be retrieved using a method called `getInstance`. [20-21 5c] [ Mark: 10 ] [ Tag: Java ]

162. Consider the following code segment. There are 4 (four) method calls in the main method. How many of them will generate compiler error and why? [20-21 5d] [ Mark: 5 ] [ Tag: Java ]

```
public class VarArgsTest {
....static void f(int... v) {
....}
....static void f(String msg, boolean... v) {
....}
....static void f(String msg, int... v) {
....}
....static void f(int n, int... v) {
....}
....static void f(double... v) {
....}
....public static void main(String[] args) {
........f("int", 10, 20, 30);
........f("boolean", true, false, false);
........f();
........f(1, 2, 3);
....}
}
```

163. What can you say about `ABC`, `S`, `X`, `Y`, and `Z` in the following statement?
`public class ABC<S extends Z & Y & X> { }`
[20-21 8d] [ Mark: 5 ] [ Tag: Java ]

164. Consider the following football clubs, their points and position in the English Premier League:

| Club | Point | Position |
|---|---|---|
| Manchester United | 90 | 1 |
| Chelsea | 80 | 2 |
| Arsenal | 75 | 4 |
| Manchester City | 30 | 19 |
| Liverpool | 25 | 20 |

Write Java code for the following:

- Declare a single Enumeration named `EPL` for the above clubs
- Write Java code to print all the clubs with their point and position. [Jan 21 4c] [ Mark: 5 ] [ Tag: Java ]

165. "If a function returns a reference of any type, then it can be used both as an l-value and r-value"- explain with an example program. [Jan 21 5b] [ Mark: 15 ] [ Tag: Theory ]

166. Write a function to check the status information of the outcome of an I/O operation using `rdstate()` function. [Jan 21 8b] [ Mark: 12 ] [ Tag: C++ ]

167. Which feature of C++ programming language do you like most and why? The feature that you mention should not be also a feature of C programming language. [Jan 20 1c] [ Mark: 10 ] [ Tag: C++, Theory ]

168. What is wrong with the following statement? [Jan 20 4a] [ Mark: 5 ] [ Tag: C++ ]

```
void func(int x=99, int y);
```

169. Differentiate between the following two statements.
(i) `int c[], x;` (ii) `int[] c, x;` **[17-18 1d]** **[ Mark: 6 ]** **[ Tag: Java ]**

170. Explain briefly dynamic method dispatching. **[17-18 3c]** **[ Mark: 6 ]** **[ Tag: Theory ]**

171. Consider the following code snippet.

```
class TestClass {
....String s;
....double d;
....public TestClass (String s, double d) (
........this.s = s;
........this.d = d;
....}
}
public class TestDemo (
....public static void main(String[] args) {
........TestClass obj1 = new TestClass("Mr. A", 18.5);
........//Your code
........TestClass obj2;
........//Your code
....}
}
```

Now write necessary Java code to write `obj1` to a file named 'test' and read the object from the same file and assign it to `obj2`. **[17-18 4c]** **[ Mark: 9 ]** **[ Tag: Java ]**

172. What is the difference between `delete` and `delete []` operators? **[17-18 6c]** **[ Mark: 3 ]** **[ Tag: C++ ]**

173. Consider the following function declaration:
`void f1 (int a=1, int b);`
Is this a valid declaration? If yes, then what does it mean? If no, then what is the problem? **[17-18 7d]** **[ Mark: 5 ]** **[ Tag: C++ ]**

174. When can a function be used both as an l-value and r-value? Explain with an example. **[17-18 8c]** **[ Mark: 6 ]** **[ Tag: Theory ]**